LAB EXPERIMENT REPORT

Experiment: User Management & RBAC in Library Book Issue System

AIM:

To configure granular User Management and Role-Based Access Control (RBAC) within a **Library Book Issue System** using Zoho Creator to secure system modules, restrict record visibility, and assign designated operational privileges for library administration, staff, and student members.

WORKFLOW:

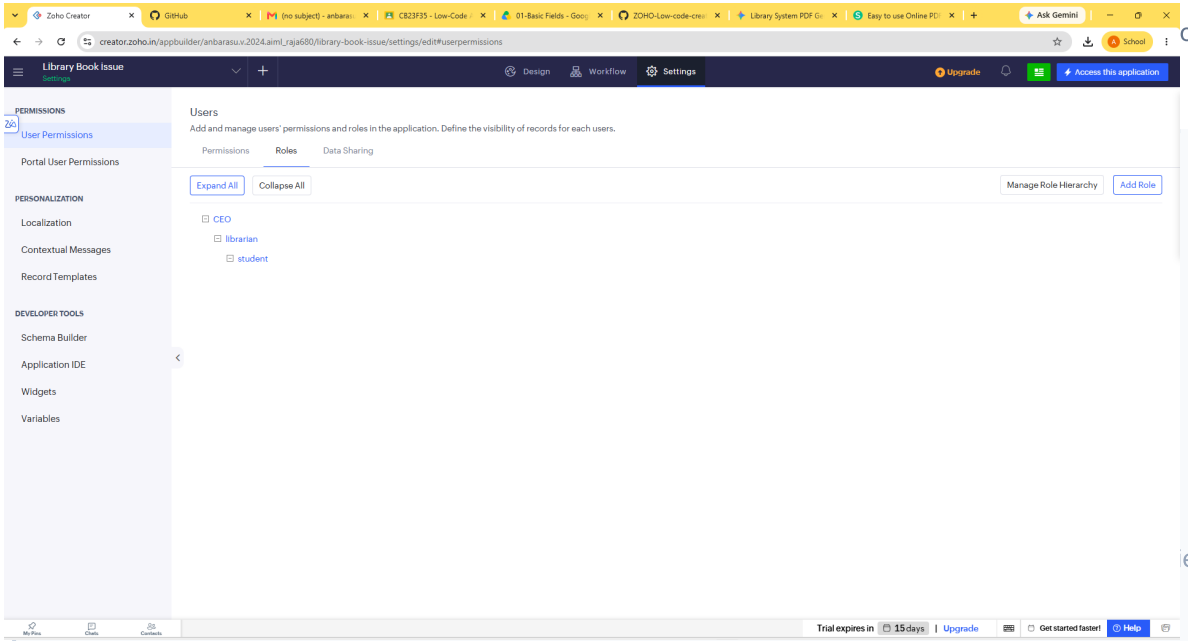
Role-Based Access Control restricts system operations and data visibility based on the user's role within the library structure. In this application, three distinct user tiers are configured to control actions on book inventory, issue registers, and fine payments:

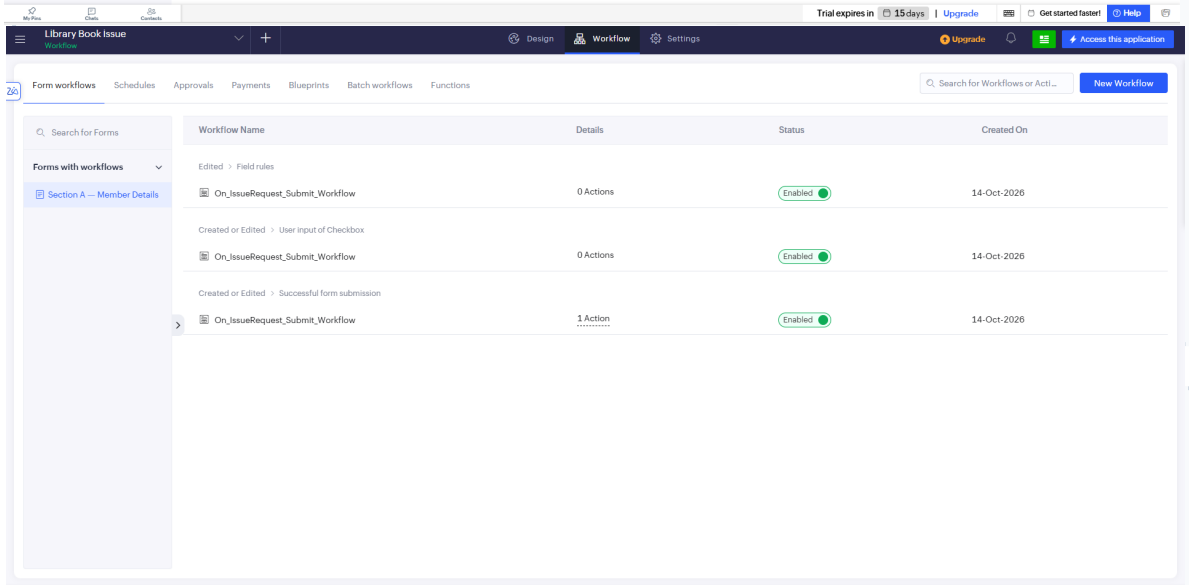
Role Name	Assigned Permissions	Data Access Scope
Chief Librarian (Admin)	Full Create, Read, Update, Delete (CRUD) & Module Configuration	All global records (Books, Members, Issue/Return Logs, Fine Overrides)
Library Assistant (Staff)	Read, Update, Issue Books, Process Returns & Collect Fines	All active Book Catalogs, Member Verification, & Branch Issue Logs
Student / Member	Create (Reserve/Request Books) & Read Own Profile	Only self-issued book history, active loans, and fine details

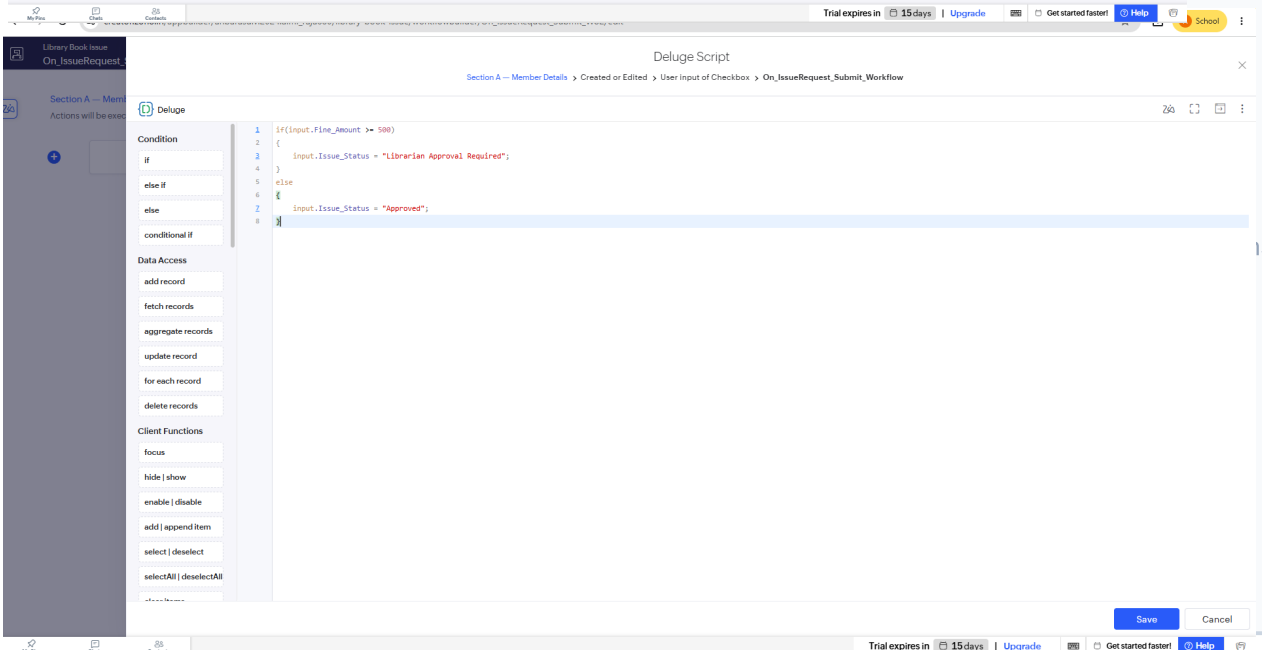
PROCEDURE:

1. **Access Developer Settings:** Log into Zoho Creator, open the *Library Book Issue System* application, and navigate to the **Settings** tab in the left-hand builder menu.
2. **Configure Roles & Hierarchy:** Navigate to **Users & Permissions > Roles**. Create custom user roles (*Library Assistant* and *Student Member*) alongside the default *Chief Librarian* role. Set reporting lines so assistants report to the Chief Librarian.
3. **Define Module Profiles:** Go to **Profiles** to configure module-level permissions for forms such as *Book Inventory*, *Book Issue Form*, and *Fine Management*. Restrict record deletion and editing rights exclusively to administrative accounts.
4. **Implement Deluge Security Workflows:** Access the **Workflow Builder** and write script checks on form submission to automatically restrict member access to admin reports and validate borrowing limits based on the user's role.
5. **Test Role Execution:** Assign sample user profiles to each role, log in as different user types, and verify that the layout, navigation tabs, and record access dynamically adjust based on assigned permissions.

SCREENSHOTS & IMPLEMENTATION PROOF







RESULT:

Verification Summary

User management and RBAC rules were successfully configured and tested in the Library Book Issue System on Zoho Creator. The implementation ensures robust data isolation, prevents unauthorized book issuing or fine modifications, and provides smooth multi-user workflow execution.